

CPRE/CYBE 230: Cyber Security Fundamentals

SAMPLE EXAM

Midterm: Thursday Feb. 29th, 2024 @ 11:00 AM

Student Name:

Student ID Number:

Lab Section (Circle one): 1 (Tuesday 4:10 PM) 2 (Thursday 12:10 PM) 3 (Friday 11:00 AM)

1. Answer the following question:

What is the network address and broadcast address of the IP address 89.55.174.15/24, and how are these addresses calculated using the given subnet mask?

2. Provide short answers to the following questions:

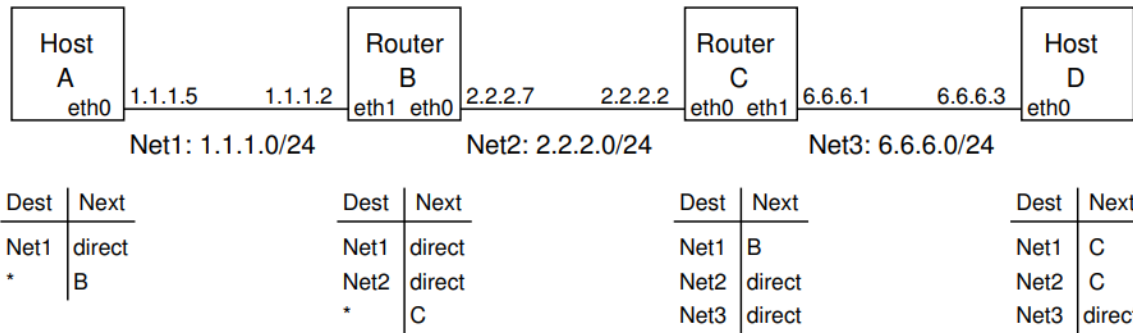
- a. What is the loop back address?
- b. Explain the following permission for the file XYZ: -rwxr--r--
- c. For Domain Name Subsystem (DNS): what is the difference between a forward look up and a reverse look up?
- d. What is the connection-oriented protocol? Give an example.
- e. How should passwords be stored in a database to follow the best practice?

3. Assuming access is limited to a Ubuntu Linux terminal with no graphical applications available, provide a single command, including the necessary options, to complete the task.

The command should not include sudo even if it is required and must be a standalone solution. The task should be completed on a computer with one LAN card, eth0, with an IP address of 1.2.3.4 and a subnet of 255.255.255.0.

- a. Display the ip address of your LAN card
- b. Display the current routing table
- c. Send ICMP echo request packets to computer with address 1.2.3.1
- d. Open the file located at /etc/hosts in a text editor
- e. Block other computers from sending packets to the webserver on your computer

4. Consider the network design diagram below.



Write a command to display the routing table of Host A

- Write the command to add the 1st row to the routing table for C.
- Write the command(s) to configure the iptables firewall on B to block all pings to/from any host on network 6.6.6.0. Even though there is only one host shown in each network, assume that there may be many more. Also assume that the default policy is ACCEPT).

5. The packets displayed below were captured using tcpdump/Wireshark (additional packets were also captured but are not shown).

No.	Time	Source	Destination	Protocol	Info
48	5.316001	0.0.0.0	255.255.255.255	DHCP	DHCP Request
50	5.318745	10.10.1.1	10.10.1.198	DHCP	DHCP ACK

Based on this information, please answer the following questions.

- Explain which computer(s) receive packet number 48. (Don't just give the destination address above)
- Explain which computer(s) receive packet number 50. (Don't just give the destination address above)
- After the above two packets have been exchanged, what is the IP address of the computer that sent packet number 48?

6. The following shows the output of the route command. Answer the questions based only on this output.

Destination	Gateway	Genmask	Flags	Metric	Ref	Use	Iface
10.10.1.0	*	255.255.255.0	U	0	0	0	eth0
default	10.10.1.1	0.0.0.0	UG	0	0	0	eth0

If a computer has a packet with the following destination addresses, where will the packet be sent? Explain your answer, by referring to the output above (e.g. which entries in the table match and why).

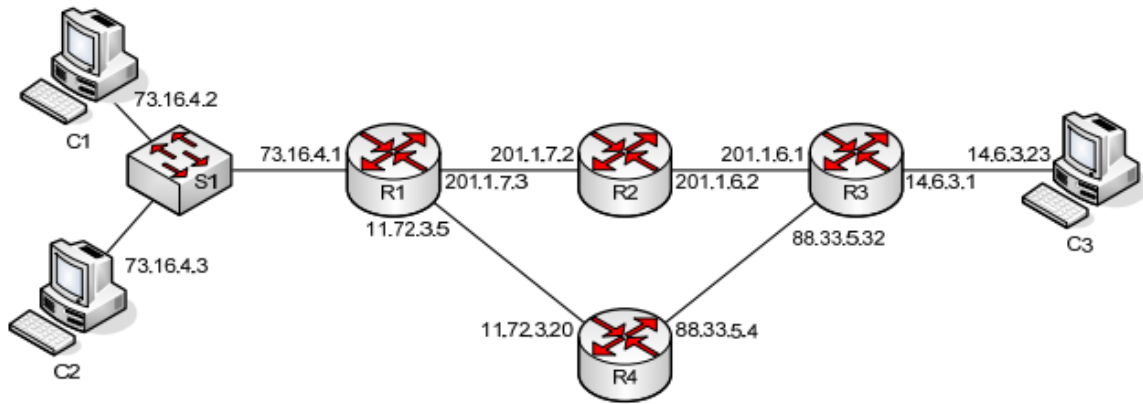
a) 10.10.1.72

b) 10.10.10.1

c) 10.10.1.1

d) 73.16.4.3

7. This diagram provides an example of an IP network with assigned IP addresses for each interface. All IP addresses are assumed to be /24, meaning the first three sets of numbers represent the network portion of the address while the last set represents the host portion of the address.



Answer the following questions based on this network.

The number of different IP networks used in this example are:

- i) 2 ii) 3 iii) 4 iv) 5 v) 6 vi) 7

Complete the routing tables for the four routers. Some entries to the tables are already given; you need to complete the remaining boxes. “Direct” means the router directly connects to the destination network. Assume minimum hop routing is used (if there are two paths with the same number of hops, then you may select one).

Router R1

<i>Destination Network</i>	<i>Next Router</i>
73.16.4.0	Direct
201.1.7.0	
11.72.3.0	
201.1.6.0	201.1.7.2
88.33.5.0	11.72.3.20
14.6.3.0	201.1.7.2

Reference Material

Common iptables Syntax

```
iptables [-t table] [-operation chain] [-p protocol] [-s srcip]  
[-d dstip] [-i inif] [-o outif] [-param1 value1 ...] -j target
```

- ▶ *table*: filter, nat, mangle
- ▶ *operation*: (first uppercase letter) Append, Delete, Insert, List, Flush, Policy, ...
- ▶ *chain*: INPUT, OUTPUT, FORWARD, PREROUTING, POSTROUTING
- ▶ *protocol*: tcp, udp, icmp, all, ...
- ▶ *srcip*, *dstip*: IP address, e.g. 1.1.1.1, 2.2.2.0/24
- ▶ *inif*, *outif*: interface name, e.g. eth0
- ▶ *param*, *value*: protocol specific parameter and value
 - ▶ sport, dport, tcp-flags, icmp-type, ...
- ▶ *target*: ACCEPT, DROP, RETURN, ...